

Module 9

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Module 9

Templates

In Module 9 we consider the topic of templates. A template is basically a code fragment with certain key pieces of information "to be named later." Effectively, when you write a template and then use it, you're making the compiler write some source code for you! Templates are a powerful mechanism for code reuse, as we'll see in several of the examples.

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Learning Objectives

- Letting the compiler write the code for you: template-izing a class
- Templates and exceptions

Lectures

Presentation

- [9.1: Functions in C](https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-9-dot-1)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-9-dot-1>)
- [9.2: Function Prototypes](https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-9-dot-2)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-9-dot-2>)

Reading

The reading for this Module is C review. Depending on your C background, you may be able to move through this material quickly.

Required Readings

- *C++ Primer Plus*, Chapters 14 and 16

Recommended Readings

- *Thinking in C++ Volume One*, pp. 573–644
- *The C++ Programming Language* (refer to as necessary)

Discussion

 Module 9 Discussion

(https://onlinelearning.berkeley.edu/courses/2291660/discussion_topics/14351904)

Assignment

 Module 9: Short Answer 9

(<https://onlinelearning.berkeley.edu/courses/2291660/assignments/33623170>)

 Module 9: Programming Assignment 9

(<https://onlinelearning.berkeley.edu/courses/2291660/assignments/33623168>)

Source Code

In each module of this course, you will be required to submit a programming assignment. For each module, I have prepared sample source code for your reference. Please take a few minutes to download and install all of these programming examples. The files have been compressed into one zip. You will need a zip utility to decompress the file.

 [sources.zip](#)

(<https://onlinelearning.berkeley.edu/courses/2291660/files/313890817/download?wrap=1>)

Course Evaluation

Please take a few minutes to participate in our End of Course Evaluation to share your opinions about this course before your course End Date. The evaluation does

not request any personal information, and your responses will remain strictly confidential. To access the evaluation, please follow the link near the bottom of the Modules tab page. You may only submit the evaluation once.